

Can triage information anticipate a hospital stay?

A patient-disjoint benchmark from target definition through explainable Gradient Boosting

The teaching question

What does a model learn, how should its score be evaluated, and what can an explanation honestly tell us?

The data contract comes before the model



Row grain

one record represents one ED stay



Availability

triage fields belong to the index-time window



Quality

missingness and derived fields remain visible

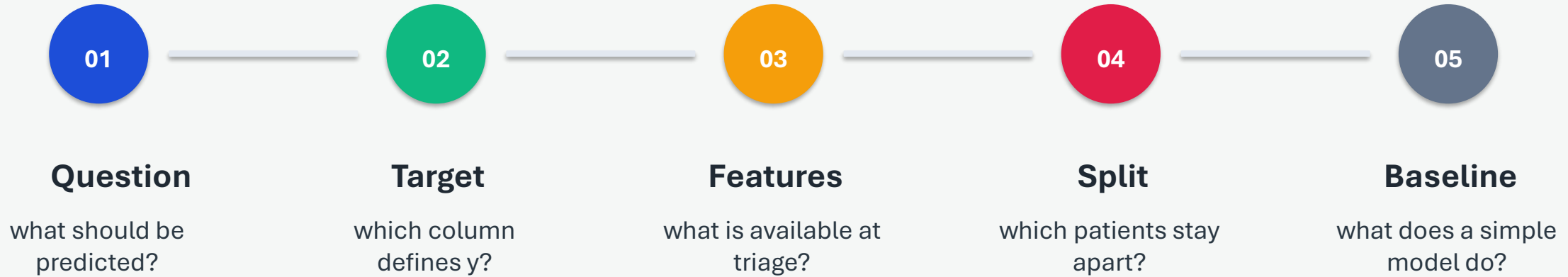


Leakage

identifiers, labels, and downstream fields stay out of X

M4 keeps the M1 contract and changes the outcome: hospitalization after triage.

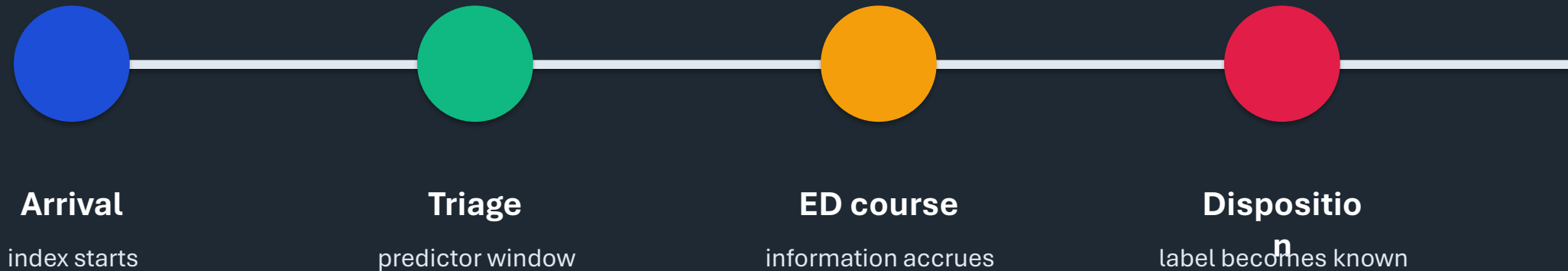
A clinical question becomes a testable ML task



The target and the time boundary decide what the score means.

Hospitalization is a later disposition label

outcome_hospitalization = 1 when hadm_id is present



The outcome is valid for evaluation, but unavailable at the moment of prediction.

The predictor boundary is a clinical assumption

Triage

vitals, acuity, pain

Context

age, gender, race, arrival
transport

Prior use

ED, hospital, and ICU
counts

Clinical flags

chief complaint and
comorbidity indicators

Included in X

Information plausibly available at triage,
transformed inside the fitted pipeline.

Excluded from X

hadm_id, disposition, later measurements,
ED departure fields, and downstream
outcomes.

The evaluation contract keeps the test population honest

9,926

held-out test stays

47.2%

hospitalized in test

5,240

non-hospitalized stays

0

patient overlap

Why patient-disjoint matters

- Repeated stays from one subject cannot teach and test the same person.
- GroupShuffleSplit uses `subject_id` to keep patients together.
- The test set estimates generalization to unseen patients from this source.

Why prevalence matters

- A positive prediction is not automatically a rare-alert event in M4.
- PPV and NPV depend on the evaluated population's prevalence.
- Do not transfer these rates unchanged to another hospital or workflow.

One preprocessing contract, four model families

Prior dummy

uses prevalence only

reference floor

Logistic regression

linear signal after preprocessing

transparent baseline

Random forest

many trees averaged

nonlinear ensemble

Gradient boosting

sequential trees correct earlier errors

explainability target

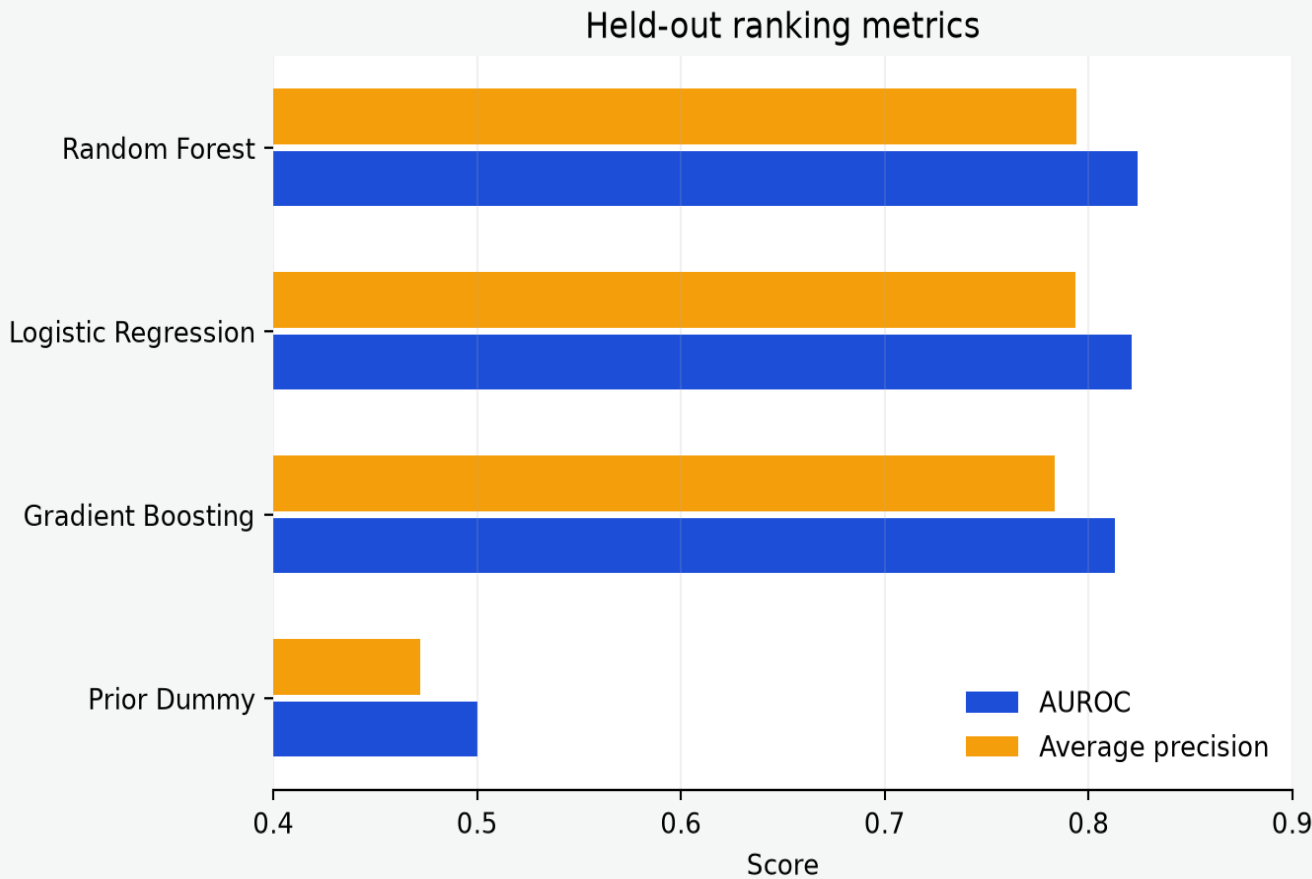
Preprocessing: imputation + missingness indicators + scaling/encoding, fitted only inside each training pipeline.

Metric vocabulary before metric results

Metric	View	Plain-language meaning
AUROC	ranking	How well positives rank above negatives across cutoffs
Average precision	ranking	Precision-recall summary, informative for the positive class
Sensitivity	threshold	Among hospitalized stays, fraction flagged positive
Specificity	threshold	Among non-hospitalized stays, fraction left negative
PPV	threshold	Among positive predictions, fraction hospitalized
NPV	threshold	Among negative predictions, fraction not hospitalized

AUROC and average precision describe ranking; the threshold metrics describe one operating point.

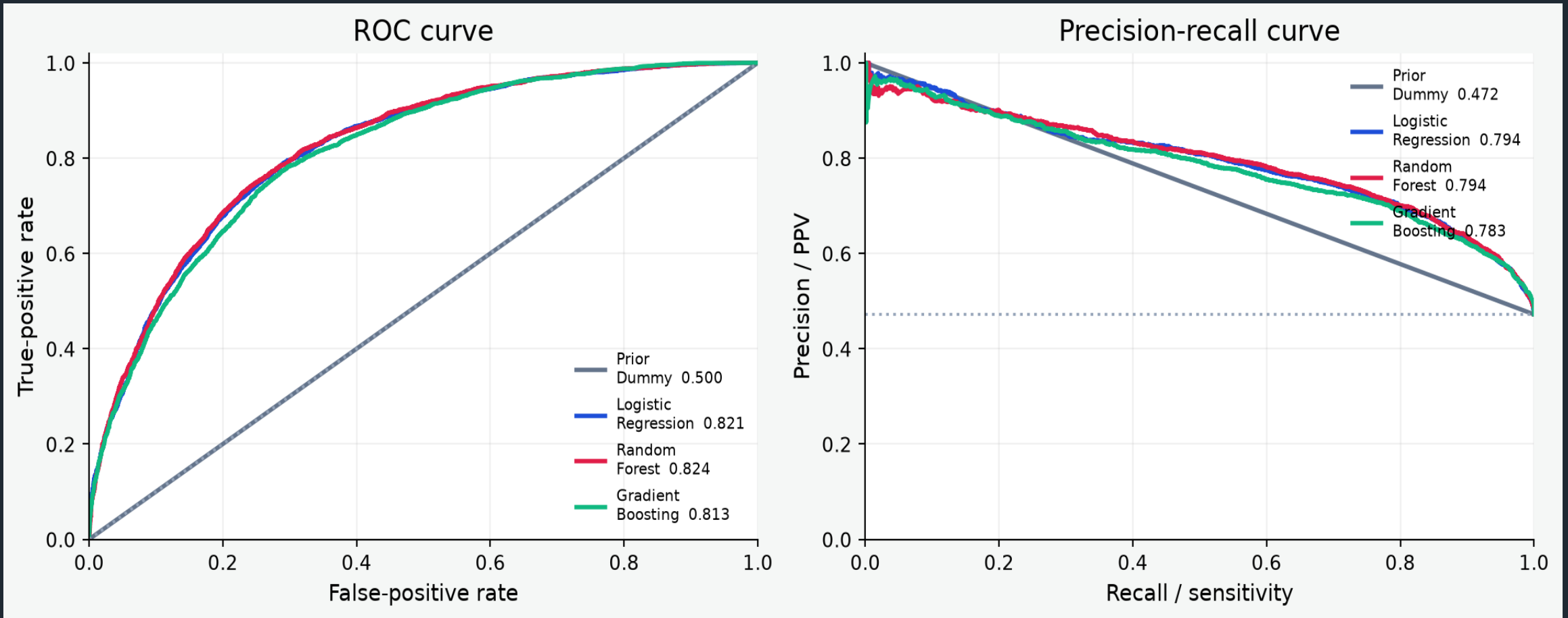
Model comparison: Random Forest ranks highest, GB remains interpretable



Model	AURO C	Avg. precision	Sensitivit y	PPV
Random Forest	0.824	0.794	0.773	0.715
Logistic Regression	0.821	0.794	0.745	0.726
Gradient Boosting	0.813	0.783	0.744	0.716
Prior Dummy	0.500	0.472	0.000	0.000

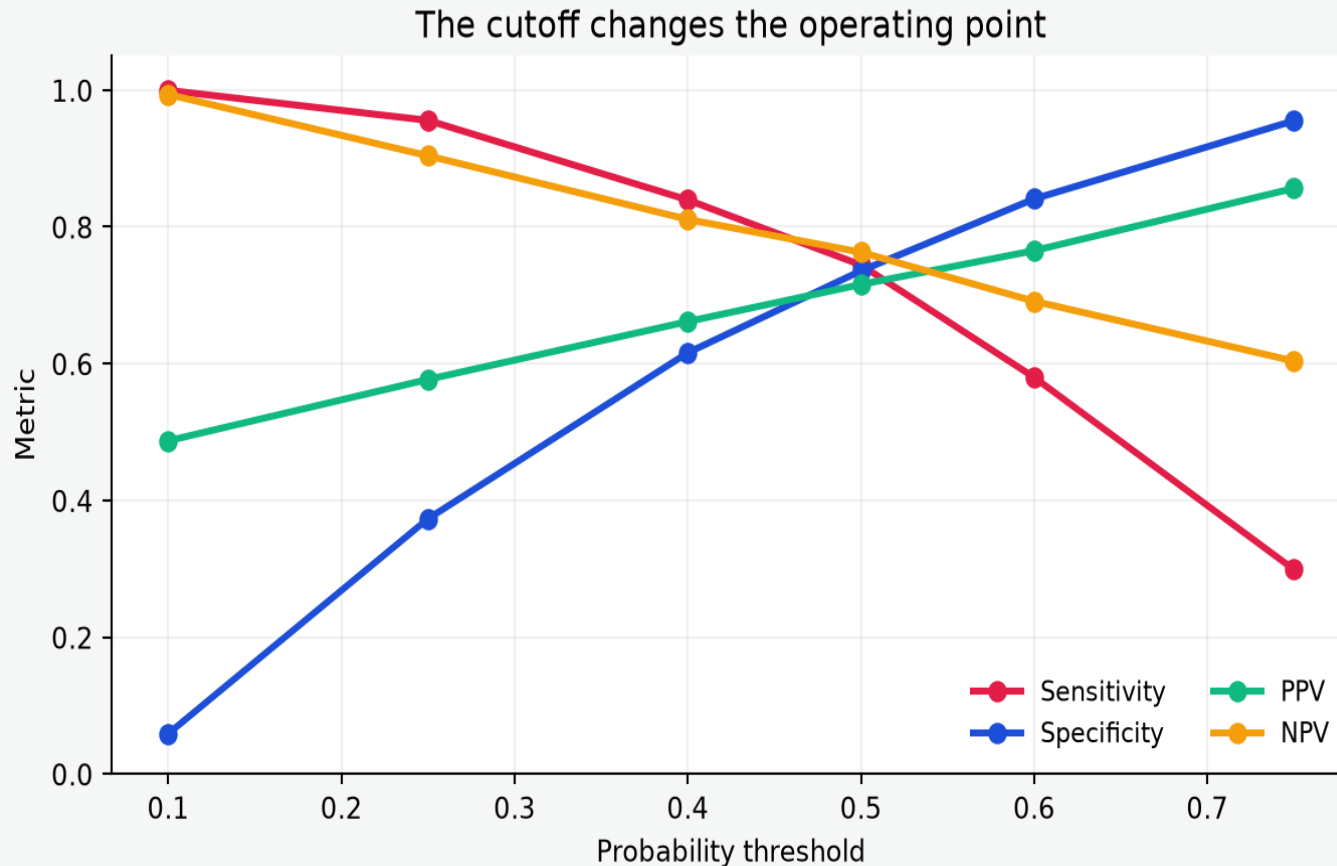
Best AUROC: Random Forest at 0.824
GB AUROC: 0.813

ROC and precision-recall curves answer different questions



ROC: ranking across false-positive rates | PR: positive-class precision as recall changes

A probability becomes an action at a cutoff



Cutoff	Sens.	Spec.	PPV	FPs
0.25	0.956	0.373	0.577	3,285
0.50	0.744	0.736	0.716	1,385
0.75	0.299	0.955	0.856	235

Lower cutoff: catch more
Higher cutoff: alert less

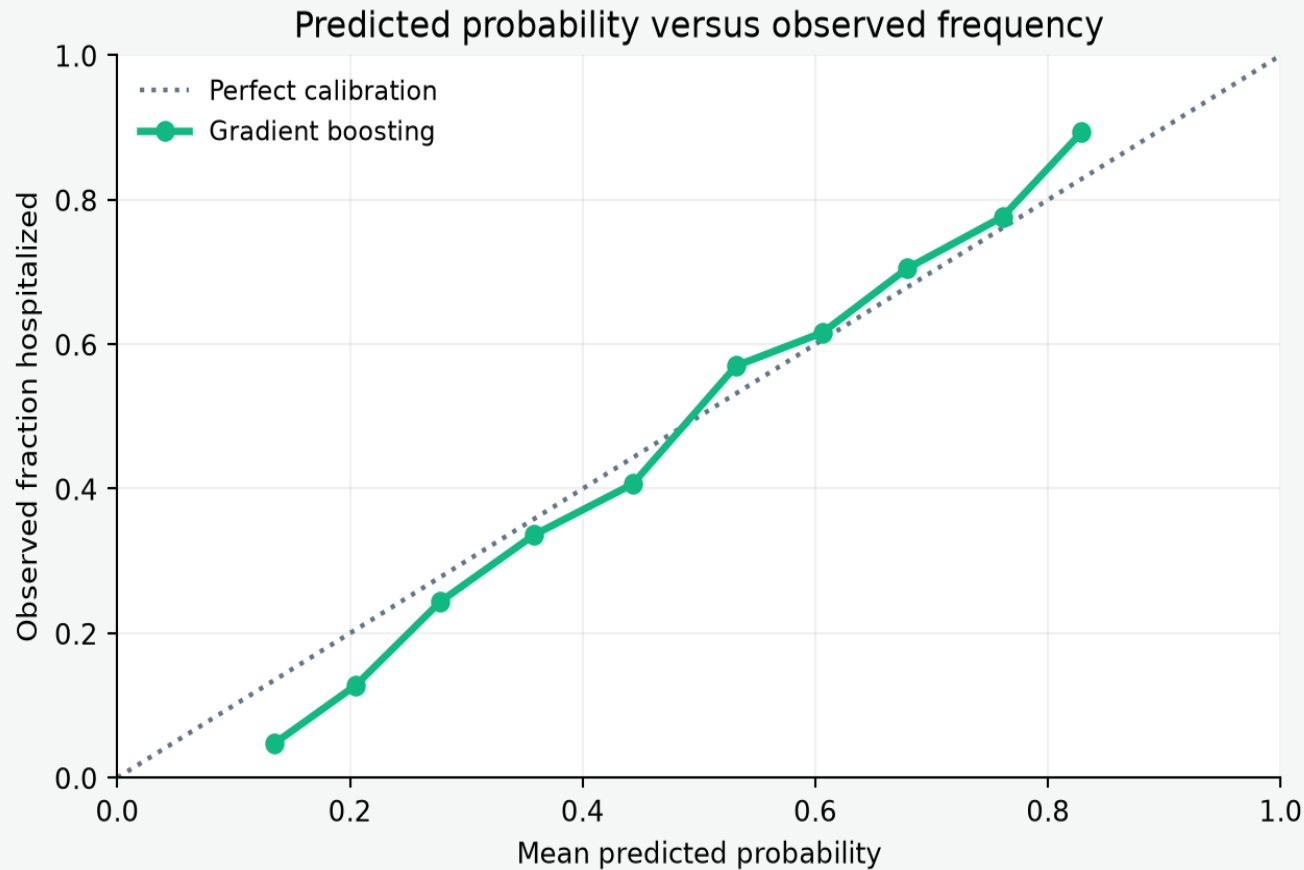
The confusion matrix counts the work created by a cutoff

	Predicted no	Predicted yes
Observed no	3,855	1,385
Observed yes	1,199	3,487
	True negative	False positive
		False negative

At threshold 0.50

- 3,487 hospitalized stays flagged
- 1,199 hospitalized stays missed
- 1,385 non-hospitalized stays flagged
- 3,855 non-hospitalized stays left negative

Discrimination is not the same as probability quality



How to read it

- A calibrated 0.70 means about 70% hospitalized among similar scored cases.
- Below the diagonal means predicted probability is higher than observed frequency. The curve is below the diagonal at lower scores, then approaches and exceeds it at higher scores.
- Calibration can shift with prevalence, hospital, time, and workflow.

Grouped cross-validation checks variation inside training

0.823

mean grouped-CV AUROC

+/- 0.002

fold-to-fold SD

What this supports

- Patients remain grouped within each fold.
- The folds show similar AUROC in this training partition.
- The test partition remains reserved for the final reported comparison.

Fold	AUROC
1	0.822
2	0.822
3	0.826

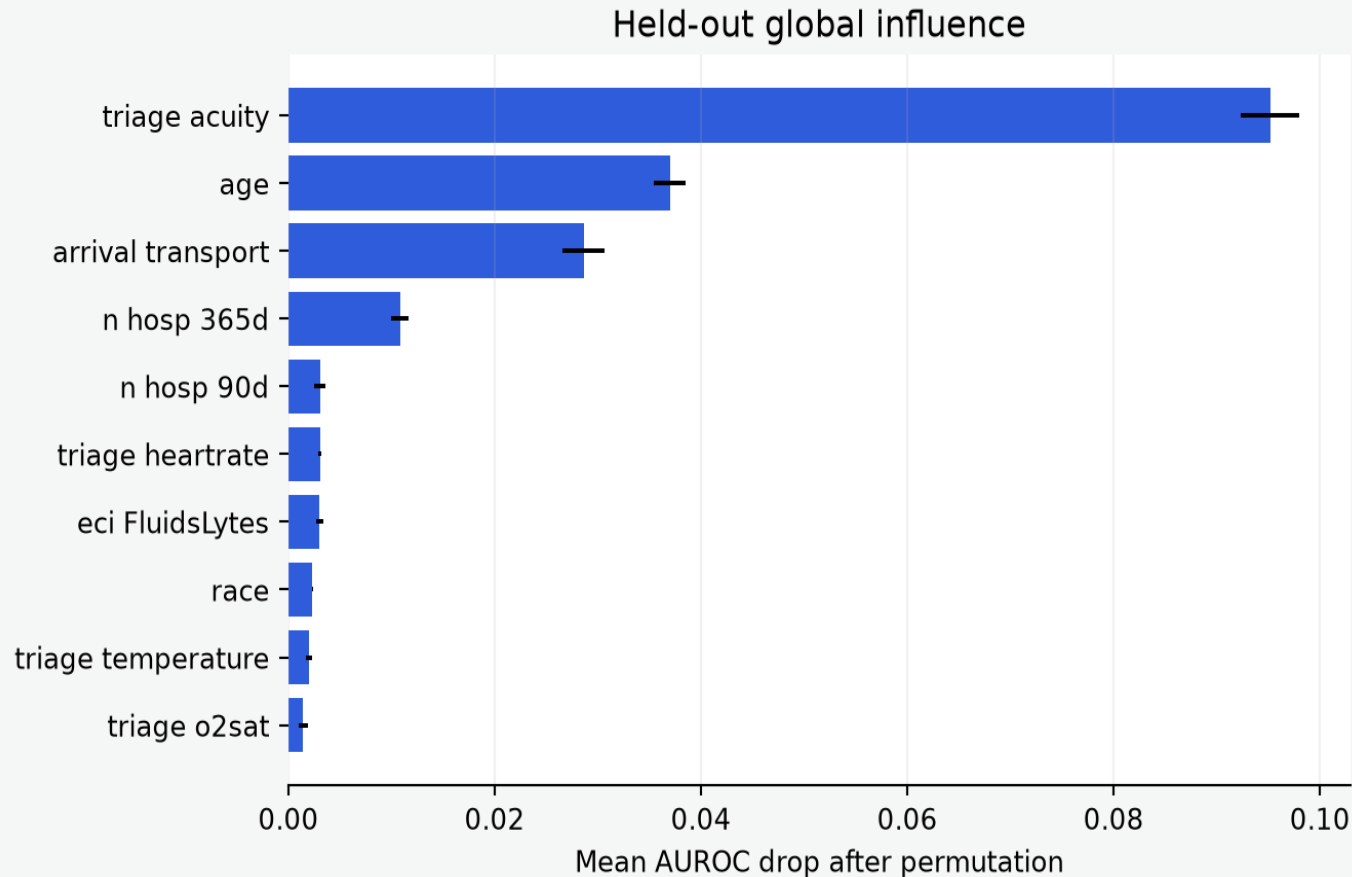
What this does not support

External validity or temporal transport.

Clinical utility, fairness, or a safe alert policy.

Confidence intervals for every subgroup or threshold.

Permutation importance asks: what happens when information is disrupted?



Permutation logic

- Shuffle one raw input in held-out rows.
- Re-score the complete fitted pipeline.
- Measure the AUROC drop.
- A larger drop means the model used that input's information on these rows.
- Correlated inputs can share or mask influence.

Three lenses, three different questions



Permutation importance

held-out AUROC drop

Does the full pipeline lose ranking information when this raw input is shuffled?



Native GB importance

tree split contribution

How much did this transformed feature contribute to the fitted tree partitions?



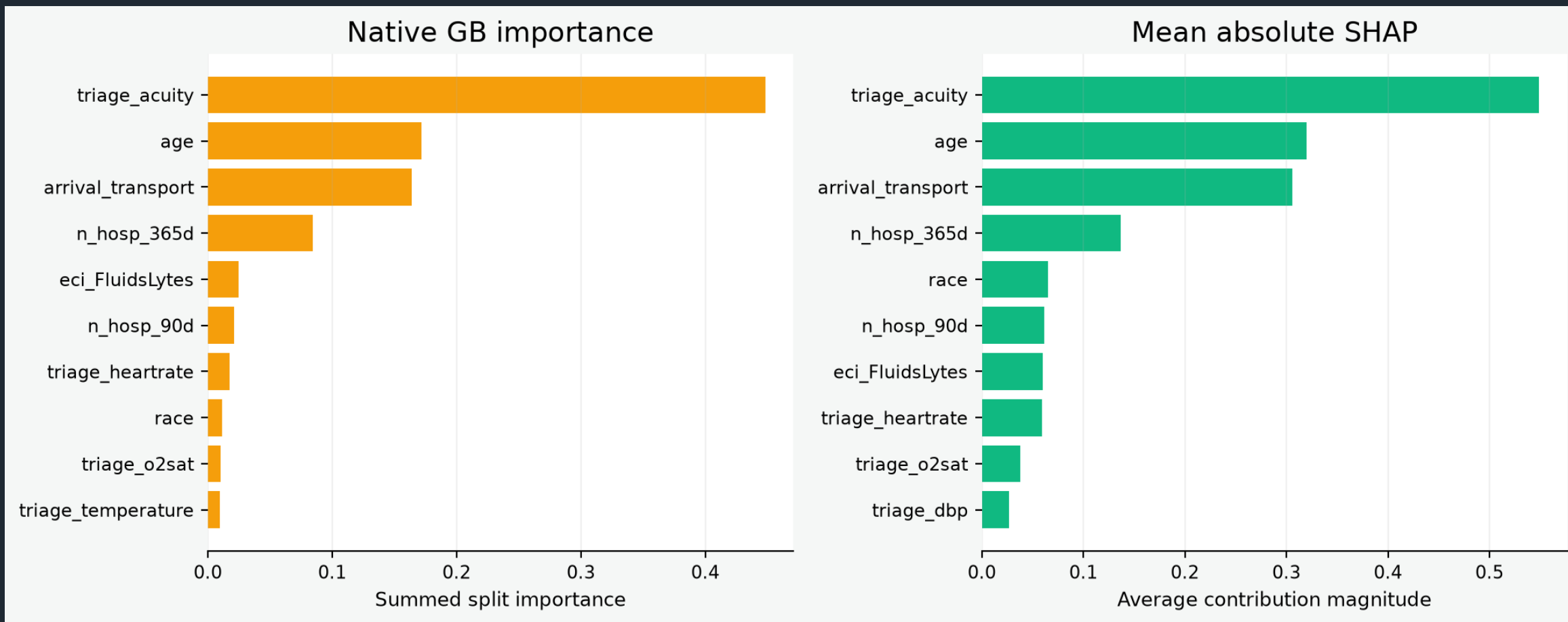
SHAP value

per-row contribution

For this row, how does this feature move the model output relative to its baseline?

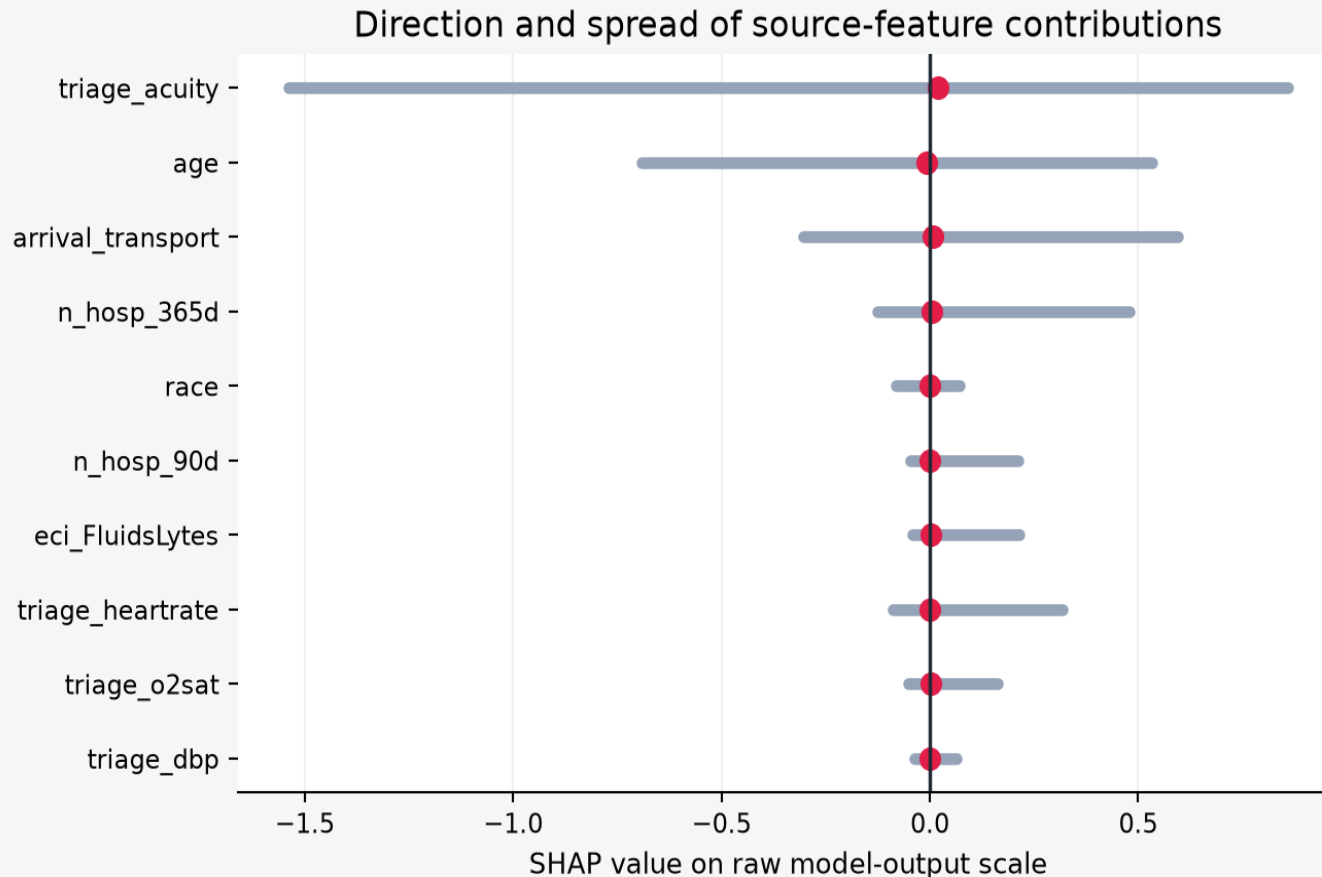
All three are model explanations. None is a causal effect estimate.

Gradient Boosting: native splits and SHAP agree on the leaders



Triage acuity, age, arrival transport, and prior hospitalization history dominate both views.

SHAP direction: positive pushes toward hospitalization



How to read a SHAP value

- Positive: moves the model output toward hospitalization.
- Negative: moves the model output away from hospitalization.
- Mean absolute SHAP: typical contribution magnitude.
- The gray range: observed contribution spread in the held-out sample.
- These values are on the estimator's raw output scale, not percentage-point probability changes.

Local explanations show why one row received its score

Row	Observed	P(hosp.)	Largest positive contributions	Largest negative contributions
21851	Yes	0.551	triage_acuity (+0.645); arrival_transport (+0.369); race (+0.055)	age (-0.575); n_hosp_365d (-0.067); triage_heartrate (-0.063)
13732	No	0.223	triage_heartrate (+0.019); chiefcom_headache (+0.005); n_ed_365d (+0.002)	triage_acuity (-0.363); arrival_transport (-0.267); age (-0.127)
1752	No	0.193	race (+0.055); triage_dbp (+0.014); chiefcom_headache (+0.005)	age (-0.485); triage_acuity (-0.350); arrival_transport (-0.259)
32802	No	0.576	arrival_transport (+0.538); age (+0.506); triage_dbp (+0.014)	triage_acuity (-0.320); race (-0.081); triage_heartrate (-0.075)

A local explanation is a model audit trail for one row, not a clinical justification or a causal narrative.